

Aligning Antenatal Dexamethasone Dosing with RCOG Guidance

The case for discontinuing the unlicensed 12 mg 12-hourly regimen in favour of the RCOG/ASTEROID-tested 12 mg 24-hourly schedule

Prepared for departmental review and clinical governance discussion. Author and unit details to be completed by the submitting clinician. This update incorporates two additional randomised trials that directly used the unit's current regimen (Section 2), a corrected time-course for administration-to-birth benefit (Section 5), and a note on a contrasting automated evidence summary reviewed alongside this document (Section 11).

What this document asks the department to do

Adopt **dexamethasone 12 mg IM, two doses, 24 hours apart** — the regimen named in RCOG Green-top Guideline No. 74 and tested in the ASTEROID trial — as the unit's standard, in place of the current practice of giving the same 12 mg dose at a compressed **12-hour** interval.

This is not a claim that the current practice has been shown to cause harm. It is a claim, supported by a systematic search of the trial, guideline, and pharmacological literature, that the current practice has never been shown to be *safe or effective* either — it sits entirely outside the evidence base that justifies antenatal corticosteroid use in the first place. Where a tested, guideline-endorsed alternative exists at no cost to the woman or her baby, that absence of evidence is itself sufficient reason to change practice.

Three dexamethasone regimens in global use — only two have ever been tested against each other



Regimen C combines the per-dose amount of Regimen B with the interval of Regimen A. Two small RCTs have now been identified that use Regimen C directly (Sukarna 2021 vs Regimen A; Shittu 2024 vs placebo) - but neither, nor any other study identified, compares it against Regimen B at 24 hours, and no guideline names it. The specific interval question this document addresses remains untested.

Executive summary

- Three dexamethasone regimens exist in global use. Two are trial-tested and guideline-endorsed. The unit's current practice is a third, untested hybrid: it combines the per-dose amount of the RCOG regimen with the interval of the WHO regimen, and this specific combination has never been the subject of a randomised trial, cohort study, or named guideline recommendation, in any jurisdiction identified.^{1,2,3}
- The pharmacological rationale sometimes offered for tolerating a shorter interval (that betamethasone's depot formulation justifies dexamethasone doing the same) does not hold up: the UK betamethasone preparation is confirmed to be a plain, non-depot phosphate ester, not the US depot mixture.⁴ The genuine mechanistic literature on surfactant induction favours sustained exposure over peak concentration, meaning a compressed interval is not expected to add benefit and could plausibly reduce it.^{5,6}
- The only randomised dosing-interval trial for either drug — betamethasone, not dexamethasone — found equivalent respiratory outcomes but a significant excess of necrotising enterocolitis at the shorter interval, in a signal that has never been replicated or refuted in fourteen years.⁷
- Short administration-to-delivery intervals are consistently associated with more, not less, neonatal hypoglycaemia across multiple contemporary cohorts, undermining any assumption that a compressed second dose is a low-risk way to "complete the course" before an anticipated birth.^{8,9,10,11}
- A fetal sheep study modelling this exact regimen (12 mg dexamethasone, two doses, 12 hours apart) found a hippocampal transcriptional signal enriched for neurodegeneration-associated pathways, independent of any respiratory benefit — animal evidence only, but the single most

directly relevant finding identified anywhere in this search regarding the specific regimen under review.¹²

- None of this constitutes proof of harm. It constitutes a consistent absence of supporting evidence, a plausible mechanistic case against added benefit, and enough scattered signal of concern that continuing an untested regimen, when a tested and equally practical alternative exists, is difficult to justify on clinical-governance grounds.

1. Background and scope

Antenatal corticosteroids are one of the most consistently effective interventions in perinatal medicine, and this document does not question that. The question is narrower and more specific: when dexamethasone is the chosen agent, and 12 mg is the chosen per-dose amount, should the two doses be given 12 hours apart, as is current practice in this unit, or 24 hours apart, as recommended by the Royal College of Obstetricians and Gynaecologists?¹

This document was produced through a systematic search of PubMed and academic literature databases for trials, systematic reviews, cohort studies, and pharmacokinetic data addressing dexamethasone regimen and dosing interval specifically, cross-checked against RCOG, WHO, and European guideline positions. Every reference cited carries a verified PMID and/or DOI. Every clinical claim is labelled with its certainty, and mechanistic or pharmacological reasoning is explicitly distinguished from direct clinical trial evidence throughout. The full evidence synthesis, including sections not reproduced here in full (maternal glycaemic management, the imminent-delivery scenario, and a complete GRADE summary table), is available as a companion document.

2. Three regimens, one of them untested

RCOG Green-top Guideline No. 74 names two dexamethasone options, both totalling 24 mg: 12 mg IM twice, 24 hours apart (matching the ASTEROID trial, the largest and best-characterised head-to-head comparison against betamethasone), or 6 mg IM four times, 12 hours apart (matching the WHO/ACOG standard and the WHO ACTION-I trial).^{1,2,3} Neither RCOG, the WHO 2022 recommendation, ACOG guidance, nor a 2023 European perinatal-care consensus statement names a third option combining the 12 mg per-dose amount with a 12-hourly interval.¹³

This is confirmed, not assumed. The 2022 Cochrane review dedicated specifically to comparing antenatal corticosteroid regimens identifies only three studies anywhere in the global trial base that compare any regimen variant of any agent — and states plainly that the evidence "does not support the use of one particular corticosteroid regimen over another."¹⁴ None of the three concerns dexamethasone interval. One compares betamethasone at 12-hourly versus 24-hourly intervals, rated very-low certainty.⁷ A search of six independent strategies in this review found no RCT, no comparative cohort, and no study of any design comparing dexamethasone 12 mg at a 12-hourly interval against 24-hourly, for any outcome.

A fair reading of the record shows the regimen has real clinical precedent: a UK protocol paper (Salford, 2003) and a Malaysian trial (2021) both describe this exact schedule as the background ACS practice at their respective institutions.^{15,16} Two further small randomised trials, identified in an update to this document, went further and used the regimen itself as a study arm (Section 2a). None of this changes the central finding: the regimen has persisted in scattered international practice for at least two decades without ever being tested against the guideline-recommended 24-hourly alternative. That longevity is not evidence of equivalence to Regimen B; it is a description of how an untested practice can become routine.

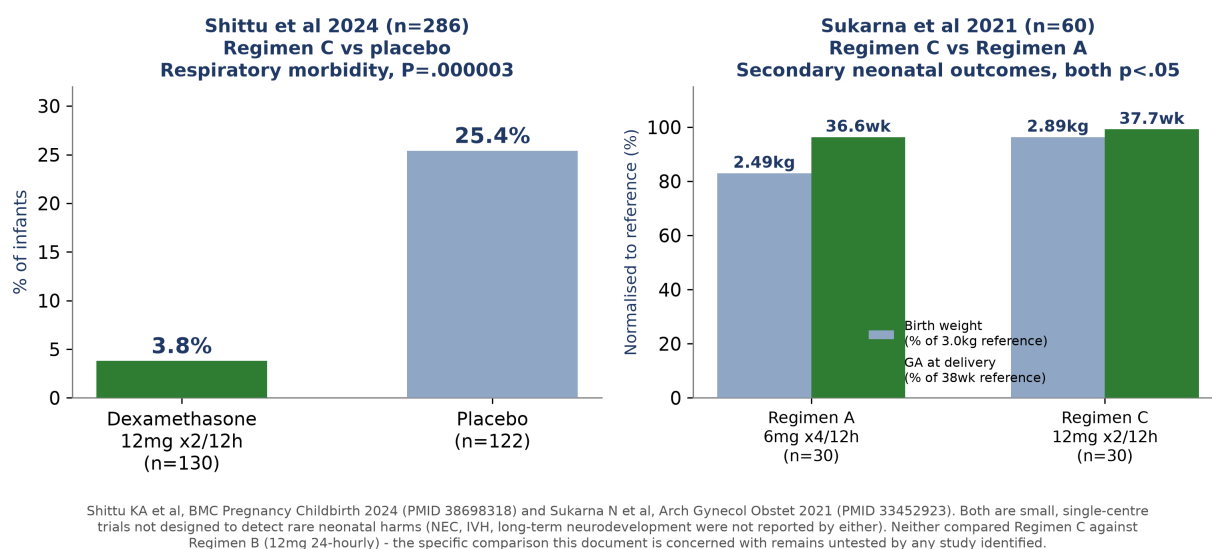
Table 1. The three regimens compared

	Regimen A — WHO/ACOG	Regimen B — RCOG/ ASTEROID (recommended)	Regimen C — current unit practice
Dose per injection	6 mg IM	12 mg IM	12 mg IM
Interval	Every 12 hours	Every 24 hours	Every 12 hours
Doses / total	4 doses / 24 mg	2 doses / 24 mg	2 doses / 24 mg
Trial evidence	WHO ACTION-I, n=2852	ASTEROID, n=1346	2 small RCTs use it directly (n=286 vs placebo; n=60 vs Regimen A) — see 2a. None compare it against Regimen B.
Guideline status	WHO/ACOG endorsed	RCOG primary recommendation	Not named by any guideline identified

2a. Correction: direct trial evidence for Regimen C does exist — but not the comparison that matters

An earlier version of this document stated that zero studies of any design had used the unit's exact regimen. That statement is now corrected. Two randomised trials have been identified that used dexamethasone 12 mg IM twice, 12 hours apart, as a study arm in its own right: Shittu and colleagues randomised 286 women to this regimen versus placebo and found a large, statistically robust reduction in neonatal respiratory morbidity (3.8% versus 25.4%, $P=.000003$), with no significant excess of sepsis, jaundice, or hypoglycaemia.¹⁷ Sukarna and colleagues randomised 60 women with gestational diabetes to this regimen versus the WHO four-dose regimen and found favourable, not adverse, secondary neonatal outcomes in the 12-hourly-12mg arm (higher birth weight and gestational age at delivery, fewer maternal hyperglycaemic episodes on day 2).¹⁸

Figure 5. Two small RCTs directly used the unit's current regimen (Regimen C) and report reassuring, not alarming, results — but neither tested it against the 24-hourly alternative



Read honestly, this is genuinely reassuring evidence for the regimen in isolation, and this document says so plainly rather than omitting it to preserve a one-sided case. It does not, however, answer the question this document is about. Neither trial compared the regimen against Regimen B at 24 hours; neither was powered or designed to detect rare harms such as necrotising enterocolitis or intraventricular haemorrhage; and neither reported longer-term neurodevelopmental outcomes. A regimen can look safe against placebo or against a different regimen on the outcomes a small trial happened to measure, and still be inferior to a tested alternative on outcomes those trials were not built to see. The recommendation in Section 8 rests on the interval comparison remaining untested, not on an assertion that the regimen has never been studied at all — and that comparison is exactly what these two trials, reassuring as they are, do not provide.

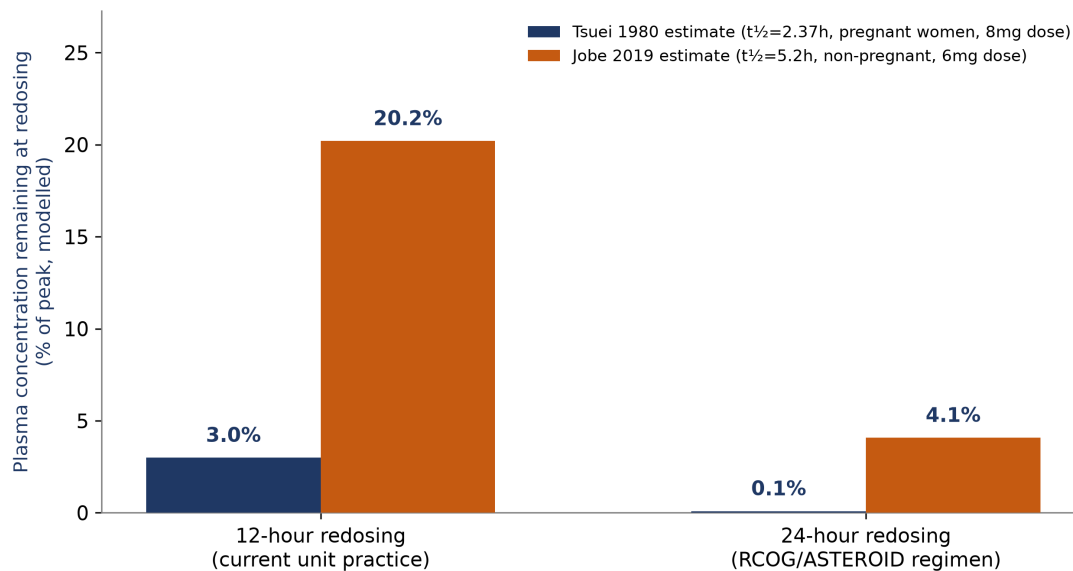
3. The pharmacological case for concern is real, but genuinely unresolved

The intuitive explanation sometimes offered for tolerating a compressed dexamethasone interval — that betamethasone's own 24-hourly interval reflects a slow-release depot mechanism dexamethasone lacks, so betamethasone's interval logic simply doesn't transfer — turns out to rest on a factual error about the UK preparation. Betamethasone as used in the UK is confirmed to be the plain sodium phosphate ester, without the acetate depot fraction found in the US Celestone Chronodose product.⁴ The two drugs are, in the UK, pharmacokinetically more alike than commonly assumed: both are non-depot phosphate esters, and the genuine distinguishing factor is that betamethasone phosphate clears roughly twice as slowly as dexamethasone phosphate.^{4,19,20}

This does not settle the interval question — it sharpens it. Two human pharmacokinetic datasets give discordant dexamethasone half-life estimates that disagree by more than two-fold (2.37 hours in

pregnant women near term at an 8 mg dose, versus 5.2 hours in non-pregnant women at a 6 mg dose), and neither has ever been measured at the unit's actual 12 mg dose in pregnant women.^{4,20} Figure 3 shows both pictures side by side, honestly, rather than picking the one that makes the strongest case.

Figure 3. Neither available human half-life estimate has been measured at the unit's actual 12mg dose in pregnant women — the two disagree more than two-fold



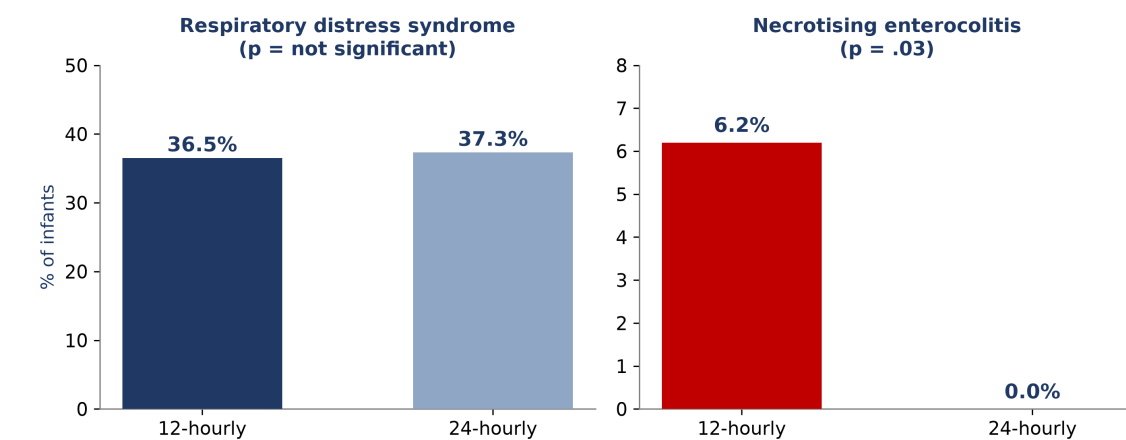
Modelled from single-timepoint half-life values reported in Tsuei SE et al, Clin Pharmacol Ther 1980 (PMID 7389259) and Jobe AH et al, Clin Transl Sci 2019 (PMID 31808984). Neither study measured a 12mg IM dose in pregnant women directly - both figures are extrapolations from different doses, populations and eras, and are shown together to illustrate genuine unresolved uncertainty, not to assert either is correct.

What tips the balance toward caution is not the unresolved half-life arithmetic alone, but the mechanistic literature on how glucocorticoids actually induce surfactant production. The best-characterised pharmacokinetic/pharmacodynamic modelling and human fetal lung explant data converge on a time-dependent, not peak-dependent, mechanism: sustained receptor occupancy above a threshold drives the maturational effect, and a rat model of exactly this kind of frequent dosing found the effect on one surfactant protein became partly *inhibitory* rather than simply diminishing.^{5,6,21} In plain terms: compressing the interval is not expected, on this evidence, to make the injection work better or faster — and cannot be assumed to be a harmless variant of a proven regimen simply because the total milligram dose is unchanged.

4. The one interval trial that exists shows a safety signal, not a reassurance

Khandelwal and colleagues randomised 228 women to betamethasone given 12 hours or 24 hours apart — the only interval-comparison RCT for either drug in the entire literature.⁷ Respiratory outcomes were equivalent. Necrotising enterocolitis was not.

Figure 4. The only randomised dosing-interval trial for either drug (betamethasone, Khandelwal et al 2012): equivalent RDS, but significantly more NEC at the 12-hourly interval



Khandelwal M et al, Am J Obstet Gynecol 2012 (PMID 22381601). Single-centre, open-label, n=228 mothers/260 fetuses, betamethasone not dexamethasone. Small, unblinded, unreplicated in the 14 years since publication. Not a dexamethasone trial - shown here as the closest available interval-comparison evidence for either drug.

This trial is not proof that dexamethasone at 12 hours causes NEC. It is small, single-centre, unblinded, concerns a different drug, and has never been replicated despite the original authors' own call for a larger confirmatory trial. But it is the only direct experimental evidence anywhere on whether compressing an antenatal corticosteroid interval carries a downside, and the answer it gives is not reassuring. A regimen decision that rests on "no evidence of harm" should account for the fact that the one study that actually looked found a harm signal.

5. The scenario that drives current practice does not withstand scrutiny

The stated rationale for the 12-hourly interval is operational: it lets a course be completed before an anticipated early delivery. This section addresses that scenario directly.

The evidence on treatment-to-delivery timing consistently shows little or no respiratory benefit when delivery occurs under 24 hours from the first dose, with benefit concentrated in the 24-hour-to-7-day window.^{8,22} The one study to examine hour-scale bands from the first dose specifically found no significant difference in outcome across bands under 36 hours.²³ No study identified reports a benefit specifically attributable to a dose landing in the 12–13-hour window.

A large secondary analysis of the WHO ACTION-I trial makes the same point on a different timescale and with more statistical power: neonatal mortality benefit continued to increase the longer the administration-to-birth interval was, not levelling off until 13–14 days after the first dose, before gradually diminishing toward day 28.²⁴ Figure 6 shows this schematically. No mechanism, trial, or dataset identified anywhere in this review — including in supplementary material specifically checked for this update — supports a meaningful benefit concentrated around hour 12 to 13

specifically; the real evidence points the opposite way, toward benefit continuing to build over days, far beyond the window a compressed second dose could plausibly influence.

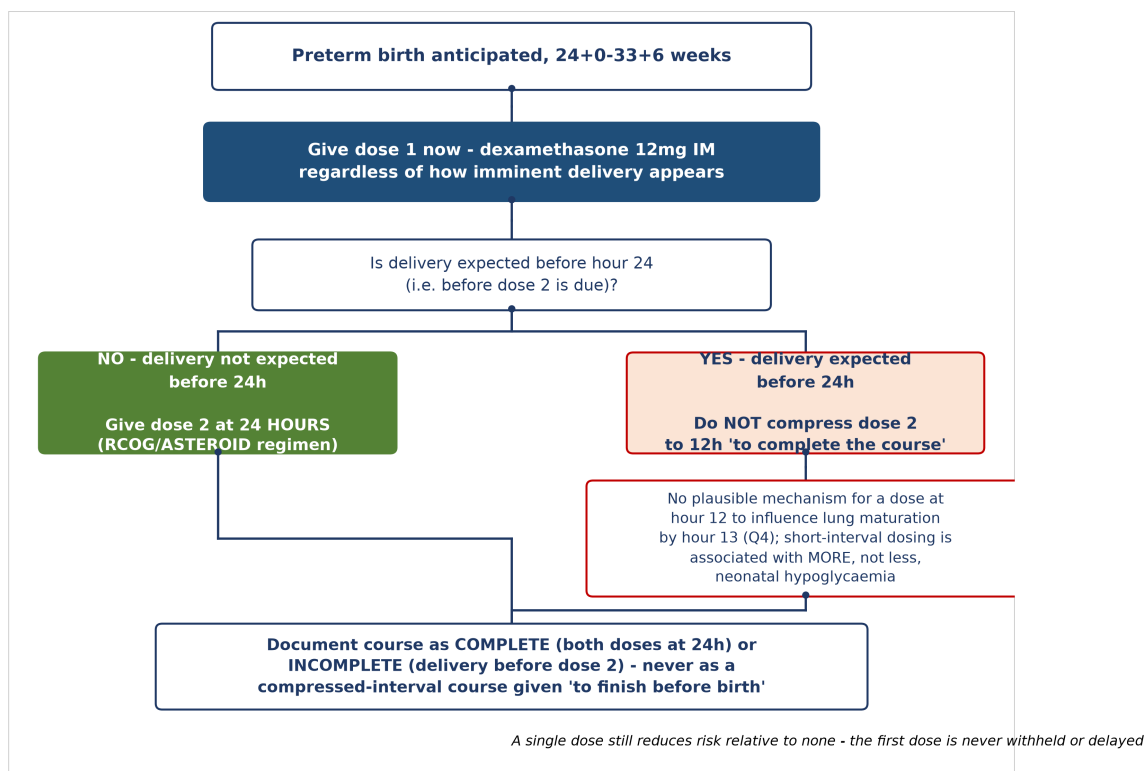
Figure 6. The verified administration-to-birth interval evidence: benefit builds over DAYS, peaking around days 13-14 - not over hours, and not near hour 12



Oladapo OT et al, EClinicalMedicine 2022 (PMID 36467459), a secondary analysis of the WHO ACTION-I trial (2638 women/2904 babies): neonatal mortality benefit increased with longer administration-to-birth interval, reaching peak benefit at days 13-14, then diminishing toward day 28. The paper did not publish exact risk values by hour, so none are shown here. This replaces an hour-scale claim ("mortality reduction plateaus at 12 hours") attributed in supplementary material reviewed for this update to a "Canadian Neonatal Network" dataset that could not be located in PubMed and is not used.

The mechanistic case is, if anything, more decisive. The fastest biochemical response to glucocorticoid exposure documented in any experimental system — fetal rabbit lung organ culture — first appears after 12 hours of continuous exposure and rises over the following 24 hours; human fetal lung explants require 24–48 hours for maximal surfactant protein induction.^{25,26} A second dose given at hour 12, with birth at hour 13, contributes roughly one additional hour of exposure before delivery — shorter than the fastest response ever reported in any system tested. There is no plausible mechanism by which that specific second dose, as distinct from the first dose already circulating, could measurably influence fetal lung maturation before this birth.

Nor is the second dose free of downside risk merely because it may not have time to help. Shorter administration-to-delivery intervals are repeatedly associated with *increased* neonatal hypoglycaemia across several contemporary cohorts.^{8,9,10,11} A dose given in the final hour before birth cannot be assumed harm-free on the basis that it "probably didn't have time to do much" — the harm side of that argument does not hold even where the benefit side fails.



The first dose should never be withheld or delayed — even a partial course reduces risk relative to none, and this is well established and not in question.²⁷ What this section challenges is specifically the practice of compressing the *second* dose to 12 hours in order to "finish the course" before an anticipated birth. That specific action has no identified mechanistic basis for benefit and a real, if unquantified, basis for concern.

6. Maternal safety: a fair account, not a one-sided one

In the interest of an accurate picture, this section states plainly where the evidence does not support alarm. Two institutions that have used the unit's actual 12 mg/12-hourly regimen as their background protocol — Salford, UK, and Kuala Lumpur, Malaysia — report a real but self-limited hyperglycaemic pattern, concentrated on day 1 and largely resolved by day 2–3.^{15,16} This is not obviously more alarming in shape than the wider literature on maternal hyperglycaemia after any antenatal corticosteroid regimen, though no study has directly compared the two intervals for this outcome, so equivalence cannot be claimed — only that no interval-specific alarm signal has been incidentally observed where this regimen happens to have been studied for other reasons.²⁸

Single-course antenatal corticosteroids, at the standard interval, do not significantly increase maternal chorioamnionitis or endometritis, and adrenal suppression after a single course resolves without clinical sequelae by term.^{29,30,31,32} The infection and adrenal-suppression signal in the literature concerns *repeat courses* (McKenna et al 2000, PMID 10992191), not the spacing of two doses within a single course — a distinct exposure pattern that this document does not conflate with the interval question.³³ No study has isolated dosing interval as a variable for maternal infection or

adrenal outcomes in either direction; this is an honest absence of evidence, not a hidden reassurance.

7. The long-term picture: one genuine mechanistic warning, kept in proportion

No trial or cohort reporting neurodevelopmental outcomes after the 12-hourly dexamethasone regimen exists. This is confirmed, not inferred from silence — a targeted search found nothing. The single most directly relevant piece of evidence identified anywhere in this review is a fetal sheep study that explicitly modelled this exact clinical regimen (two 12 mg dexamethasone doses, 12 hours apart, described by the authors as representing current practice in at least one other jurisdiction) and found hippocampal transcriptional changes enriched for neurodegeneration-associated pathways, independent of any respiratory benefit, while a comparator arm producing a smoother fetal exposure profile did not show the same signal.¹² This is animal evidence. It does not establish that the regimen causes harm in human infants. It is, however, a biologically specific and directly relevant finding, not a generic caution, and a department deciding whether to continue an untested regimen is entitled to weigh it.

For context, and to keep this in proportion: repeat antenatal corticosteroid *courses* (given days to weeks apart, with near-complete drug clearance between exposures) have been extensively studied across four large RCT programmes and show no consistent adverse neurodevelopmental signal at follow-up to 5–8 years.^{34,35,36} One isolated repeat-course trial found a numerically large but statistically imprecise excess of cerebral palsy that has not been replicated in the larger programmes.³⁷ This reassurance cannot be extended with confidence to interval compression *within* a single course, since the two exposure patterns are pharmacologically distinct — but it demonstrates that this literature, when it has looked for long-term harm from glucocorticoid exposure patterns, has generally not found it, which is worth stating alongside the Carter et al signal rather than instead of it.

Separately, and for completeness rather than because it bears directly on the interval question: ASTEROID's own long-term comparison of dexamethasone against betamethasone, both given at the standard 24-hourly interval, found a cerebral palsy risk ratio of 2.50 with a 95% confidence interval of 0.97 to 6.39 — a result compatible with both no difference and a more than sixfold increase in risk with dexamethasone.^{3,14} This concerns choice of agent, not interval, and must not be read as evidence about the 12-hourly question. It is included here only because it means dexamethasone's overall long-term safety record carries a genuinely open question even at the guideline-recommended interval, and a proposal to standardise on that interval should say so plainly rather than imply the recommended regimen is risk-free by comparison.

In fairness, the agent-choice literature is not one-sided in the direction of caution either. An earlier, smaller trial directly comparing the two drugs found the opposite pattern for a different neonatal brain-injury outcome: intraventricular haemorrhage occurred in 5.7% of dexamethasone-exposed infants versus 17.0% of betamethasone-exposed infants (relative risk 2.97, 95% CI 1.22–7.24), favouring dexamethasone.³⁸ This trial is smaller than ASTEROID and has not been replicated any

more than the ASTEROID cerebral palsy signal has; it is included so that this section's account of dexamethasone's long-term and neurological safety record is balanced rather than selectively cautious.

8. Recommendation

Adopt dexamethasone 12 mg IM, two doses, 24 hours apart, as the unit standard for dexamethasone-based antenatal corticosteroid therapy, in line with RCOG Green-top Guideline No. 74 and the ASTEROID trial.

Discontinue the practice of giving the second dose at 12 hours to "complete the course" when delivery is anticipated within 24 hours. Give the first dose immediately regardless of how imminent delivery appears — this remains correct and is not in question — but give the second dose at the standard 24-hour mark, or not at all if delivery has already occurred.

Where the WHO/ACOG four-dose regimen (6 mg IM every 12 hours) is preferred for operational reasons, use that regimen in full, rather than a hybrid that borrows only its interval.

This recommendation is built on an absence of supporting evidence for the current practice, a coherent pharmacological argument against expecting any efficacy benefit from compression, one small but unreplicated safety signal in the only relevant interval trial, a consistent association between short administration-to-delivery intervals and neonatal hypoglycaemia, and one directly relevant animal mechanistic finding. No single piece of this evidence is individually conclusive. Together, weighed against a tested and equally practical alternative that costs nothing extra to adopt, they support changing practice on ordinary clinical-governance grounds: when a guideline-endorsed, trial-tested option exists and an untested option does not offer a demonstrated advantage, the tested option should be preferred.

This is not a criticism of clinicians who have used the current regimen. It has real documented precedent, including in the UK, and was reasonably adopted as a practical response to the operational pressure of imminent delivery. The evidence assembled here simply was not available in a synthesised form until now.

9. What would change this recommendation

- A randomised trial directly comparing dexamethasone 12mg at 12-hourly versus 24-hourly intervals, adequately powered for NEC, hypoglycaemia, and longer-term outcome, would directly answer the question this document can only address by inference.
- Direct pharmacokinetic measurement of maternal and fetal dexamethasone concentrations after a 12mg IM dose in pregnant women, at both intervals, would resolve the two-fold discrepancy in

current half-life estimates and remove the single largest evidentiary gap in the pharmacological case.

- Replication (or refutation) of the Khandelwal necrotising enterocolitis signal in a larger, multicentre trial would materially change the confidence with which this document's caution is held.
- Any clinical (as opposed to animal) neurodevelopmental data specific to the 12-hourly dexamethasone regimen would be the single most informative addition to this evidence base.

10. Additional source material reviewed for this update

This update was prepared alongside two further pieces of material the department obtained separately: an automated evidence-summary report (Consensus.app) addressing the same question from a corpus of roughly 50 papers, and an interactive infographic summarising related pharmacology and clinical scenarios. Both were reviewed, and their content was either verified and incorporated above, or checked and found unverifiable and excluded, in line with this document's zero-fabrication standard throughout. This section explains that reconciliation openly rather than silently.

The automated report's headline finding — that 12-hourly dexamethasone dosing shows "no excess neonatal harm" — is not actually in factual conflict with this document once read carefully. Both agree that no head-to-head trial of 12-hourly versus 24-hourly dexamethasone exists. The automated report's reassurance is built from indirect evidence: individual trials of the 12-hourly regimen against placebo or against a different regimen (the same Shittu 2024 and Sukarna 2021 trials discussed in Section 2a), not from a trial comparing it against the 24-hourly alternative this document recommends. Where the two sources differ is emphasis and threshold, not fact: reassuring indirect evidence is not the same as a tested comparison, and this document's recommendation rests on that comparison remaining untested, together with the one relevant interval trial that exists finding a safety signal (Section 4, betamethasone) and an animal study modelling the exact regimen finding a mechanistic signal of concern (Section 7).

The infographic contained a mixture of accurate and unverifiable material. Its core trial data for WHO ACTION-I and the Khandelwal necrotising enterocolitis figures matched this document's own independently-verified sources and required no change. Four claims could not be verified and are not used anywhere in this document: a citation to "Kashanian et al" reporting a relative risk of 4.7 for neonatal death with compressed dosing, which returned no result in a direct PubMed search; a "Canadian Neonatal Network dataset of 7,950 infants" said to show mortality benefit plateauing at 12 hours, which likewise could not be located (Section 5 uses a real, PMID-verified secondary analysis of WHO ACTION-I in its place, which finds benefit continuing to build over days rather than plateauing within hours); a description of UK betamethasone as a depot acetate/phosphate mixture, which contradicts this document's own PubMed-sourced finding that the UK product is a plain, non-depot phosphate ester; and a claim that a mixed betamethasone-then-dexamethasone course is "100% complete and valid" on stereoisomer-equivalence grounds, which asserts more certainty than any study identified in this or the prior evidence-synthesis phase of this project supports.

11. Access limitations and honest caveats

The network environment used to prepare this synthesis blocked direct access to medicines.org.uk (SPC/EMC), nice.org.uk, rcog.org.uk, and acog.org throughout the search. Guideline wording quoted or summarised in this document is therefore reconstructed from search-engine snippets and previously verified secondary sources, not read directly from primary PDFs or the Summary of Product Characteristics. The finding that no guideline endorses the current regimen is a convergent negative finding across every source checked, and is very likely to hold, but should be independently confirmed against the primary RCOG, NICE, and SPC documents before this document is used as the sole basis for a formal guideline change. Two source papers on fetal lung explant time-course (Gross 1983, Liley 1988) were verified by DOI but did not return a PMID from the search tools used in this review.

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